

MODEL QUESTION PAPER

TED (10) 5081

Reg. No.

(Revision - 2010)

Signature.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/

TECHNOLOGY - MARCH - 2013

BIOCHEMICAL ENGINEERING

(Maximum marks: 100)

[Time : 3 hours

(Answer all questions in one or two sentences. Each carries 2 marks)

I

- 1 What are Lipids ? Give its classification ?
- 2 Define enzyme
- 3 What is meant by aerobic fermentation?
- 4 What is chromatography?
- 5 List the reactors for immobilized enzyme systems. (5x2=10)

PART-B

II

(Answer any five questions. Each carries 6 marks)

- 1 Write short note on 6
i) Viruses ii) Algae iii) Protozoa
- 2 List the advantages of enzyme immobilization 6
- 3 Illustrate general requirements and basic components of fermentation process. 6
- 4 Explain ultra filtration. 6
- 5 What is Dialysis ? Give its application with example. 6
- 6 Draw the flow diagram of manufacture of ethyl alcohol by fermentation process? 6
- 7 Explain Entrapment process of enzyme immobilization 6

PART-C

(Answer one full question from each module. Each question carries 15 marks)

MODULE-I

III

Explain the structure of prokaryotic cell with neat sketch. 15

OR

IV Explain the structure and functions of RNA ? 10
Write a short note on Vitamins

MODULE-II

V (a) Explain Fischer Lock and Key and hypothesis in enzymes ? 10
(b) Write the properties of immobilized enzyme. 5

OR

VI Explain the chemical theories those Enzymes as biological catalyst. 10
Define active site enzyme. 5

VI (a) What are the general requirements of fermentation process ? 9
(b) Explain submerged fermentation. 6

OR

VIII Explain the production of Penicillin by fermentation? 15

MODULE-IV

IX (a) Explain affinity chromatographic process. 12
(b) List the various application of chromatographic separation process. 3

OR

X Explain different types of membrane modules ? 15